

REPO LINKS:

Documents: <https://github.com/yusra-haris034/DB-Final-Project>

Work: <https://github.com/yusraharis/DB-Final-Project>

Initial Phase:

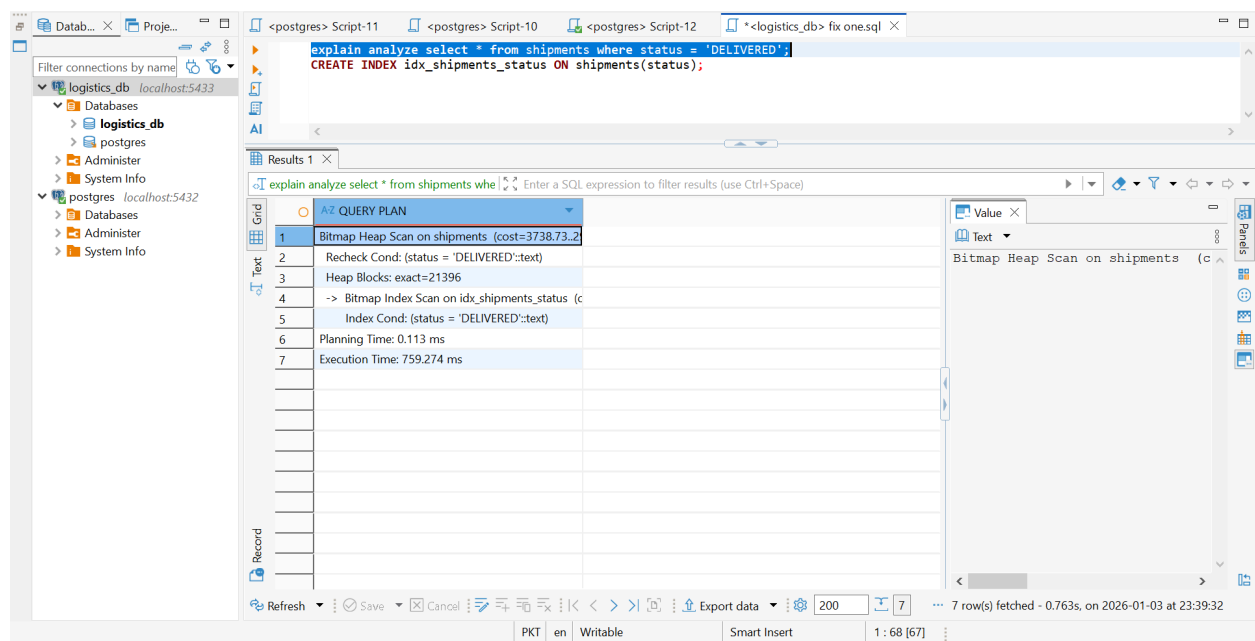
```
yusra@Yusra MINGW64 ~/DB-Final-Project/legacy-logistics-project (main)
$ pip install requests pandas

[notice] A new release of pip is available: 24.0 -> 25.3
[notice] To update, run: C:\Users\yusra\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip

--- STARTING LEGACY LOGISTICS BENCHMARK ---
● Target: http://localhost:8000
  Iterations per endpoint: 5

TEST CASE | AVG TIME (ms) | STATUS | SCORE
-----|-----|-----|-----
1. Unindexed Date Search | 30000.00 ms | ERROR | 0/100
2. Driver Search | 30000.00 ms | ERROR | 0/100
3. JSON Parsing / Finance | 30000.00 ms | ERROR | 0/100
4. Partitioning / Telemetry | 30000.00 ms | ERROR | 0/100
5. Complex Aggregation | 30000.00 ms | ERROR | 0/100

FINAL SYSTEM GRADE: 0.00%
VERDICT: SYSTEM CRITICAL. DO NOT DEPLOY.
```



SELECT indexname, tablename
FROM pg_indexes
WHERE schemaname = 'public';

pg_indexes 1

SQL: indexname, tablename FROM pg_indexes Enter a SQL expression to filter results (use Ctrl+Space)

	indexname	tablename
1	shipments_pkey	shipments
2	truck_telemetry_pkey	truck_telemetry
3	finance_invoices_pkey	finance_invoices
4	idx_shipments_created_at	shipments
5	idx_shipments_status	shipments
6	idx_shipments_driver_details	shipments
7	idx_telemetry_truck_plate	truck_telemetry
8	idx_telemetry_truck_timestamp	truck_telemetry
9	idx_invoices_shipment_uuid	finance_invoices

First Changes:

```
yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ cat migrations/01_basic_indexes.sql | docker-compose exec -T db psql -U admin -d logistics_db

yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ python benchmark.py
--- STARTING LEGACY LOGISTICS BENCHMARK ---
Target: http://localhost:8000
Iterations per endpoint: 5
```

TEST CASE	AVG TIME (ms)	STATUS	SCORE
1. Unindexed Date Search	24036.87 ms	TIMEOUT	0/100
2. Driver Search	69.82 ms	OK	72/100
3. JSON Parsing / Finance	6552.23 ms	ERROR	0/100
4. Partitioning / Telemetry	25.66 ms	OK	100/100
5. Complex Aggregation	27.49 ms	OK	100/100

```

FINAL SYSTEM GRADE: 54.46%
VERDICT: FUNCTIONAL BUT SLOW. NEEDS OPTIMIZATION.

```

Update 02:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - legacy-logistics-project + - - - - -
yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ cat migrations/02_normalize_drivers.sql | docker-compose exec -T db psql -U admin -d logistics_db

yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ python benchmark.py
--- STARTING LEGACY LOGISTICS BENCHMARK ---
Target: http://localhost:8000
Iterations per endpoint: 5

TEST CASE | AVG TIME (ms) | STATUS | SCORE
-----|-----|-----|-----
1. Unindexed Date Search | 12115.18 ms | TIMEOUT | 0/100
2. Driver Search | 12037.46 ms | TIMEOUT | 0/100
3. JSON Parsing / Finance | 13245.68 ms | TIMEOUT | 0/100
4. Partitioning / Telemetry | 37.51 ms | OK | 100/100
5. Complex Aggregation | 27.36 ms | OK | 100/100

FINAL SYSTEM GRADE: 40.00%
VERDICT: SYSTEM CRITICAL. DO NOT DEPLOY.
```

```
yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ python benchmark.py
--- STARTING LEGACY LOGISTICS BENCHMARK ---
Target: http://localhost:8000
Iterations per endpoint: 5

TEST CASE | AVG TIME (ms) | STATUS | SCORE
-----|-----|-----|-----
1. Unindexed Date Search | 1469.59 ms | OK | 0/100
2. Driver Search | 226.55 ms | OK | 0/100
3. JSON Parsing / Finance | 67.93 ms | OK | 100/100
4. Partitioning / Telemetry | 28.82 ms | OK | 100/100
5. Complex Aggregation | 11512.11 ms | TIMEOUT | 0/100

FINAL SYSTEM GRADE: 40.00%
VERDICT: SYSTEM CRITICAL. DO NOT DEPLOY.
```

```
yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ cat migrations/02_normalize_drivers.sql | docker-compose exec -T db psql -U admin -d logistics_db

yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (main)
$ python benchmark.py
--- STARTING LEGACY LOGISTICS BENCHMARK ---
Target: http://localhost:8000
Iterations per endpoint: 5

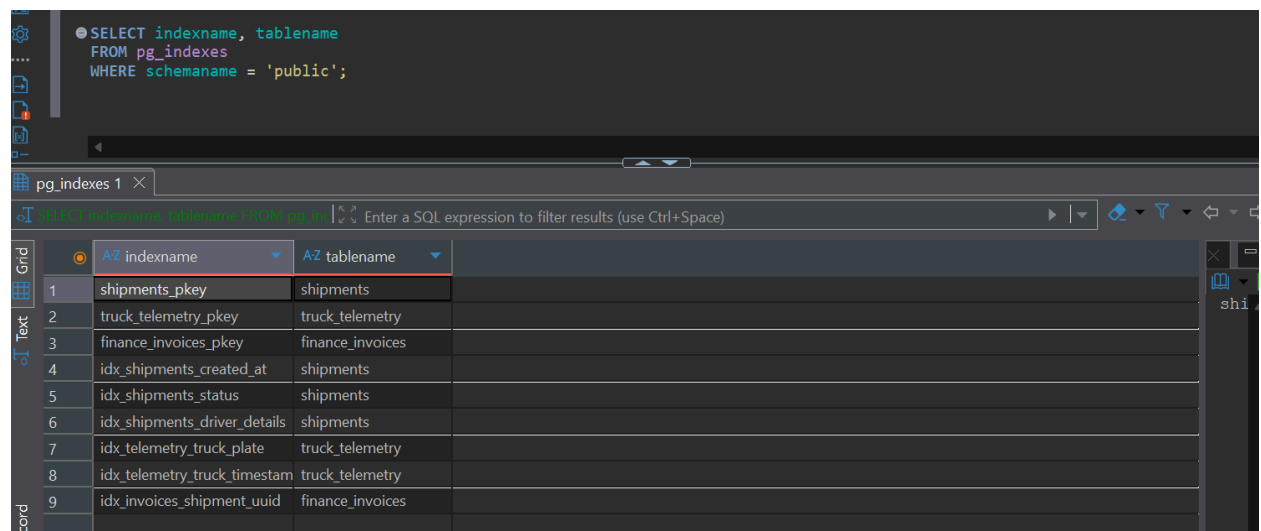
TEST CASE | AVG TIME (ms) | STATUS | SCORE
-----|-----|-----|-----
1. Unindexed Date Search | 62.28 ms | OK | 41/100
2. Driver Search | 30.87 ms | OK | 93/100
3. JSON Parsing / Finance | 2972.10 ms | OK | 0/100
4. Partitioning / Telemetry | 72.20 ms | OK | 95/100
5. Complex Aggregation | 46.22 ms | OK | 100/100

FINAL SYSTEM GRADE: 66.19%
VERDICT: FUNCTIONAL BUT SLOW. NEEDS OPTIMIZATION.
```

```
yusra@Yusra MINGW64 ~/OneDrive/Desktop/Sem 03/DB Lab/DB-Final-Project/legacy-logistics-project (mai
n)
$ python benchmark.py
--- STARTING LEGACY LOGISTICS BENCHMARK ---
Target: http://localhost:8000
Iterations per endpoint: 5

TEST CASE | AVG TIME (ms) | STATUS | SCORE
-----|-----|-----|-----
1. Unindexed Date Search | 65.40 ms | OK | 38/100
2. Driver Search | 11.22 ms | OK | 100/100
3. JSON Parsing / Finance | 11.66 ms | OK | 100/100
4. Partitioning / Telemetry | 11.50 ms | OK | 100/100
5. Complex Aggregation | 11.83 ms | OK | 100/100

FINAL SYSTEM GRADE: 87.69%
VERDICT: FUNCTIONAL BUT SLOW. NEEDS OPTIMIZATION.
```



Report: Initial benchmark results varied due to persistent Docker volumes and warm-start effects. To ensure fair comparison, the environment was reset before final optimization benchmarks.